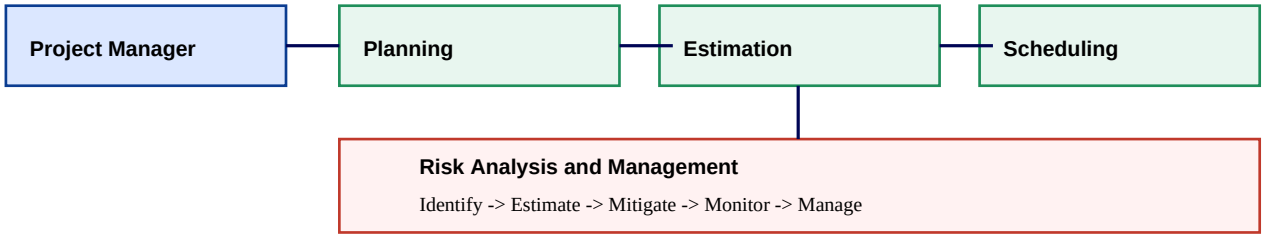
Unit 5: Software Project Management

Short Notes | Covers Every Topic | Software Engineering (DU-2304CS304)

This is a short, complete revision PDF for Unit 5. It covers all main topics: project manager responsibilities, planning, estimation, LOC, Function Points, COCOMO, scheduling, Gantt charts, PERT/CPM, and risk management.



1. Software Project Manager

A Software Project Manager completes a software project successfully within time, cost, scope, and quality.

Responsibilities

- Plan the project and set objectives.
- Divide work and assign tasks.
- Manage people, tools, money, and time.
- Identify and manage risks.
- Solve problems and make decisions.
- Track progress and ensure quality.
- Control budget and ensure on-time delivery.
- Communicate with team and customer.
- Motivate the team and resolve conflicts.

2. Software Project Planning

Project planning prepares a roadmap for the software project. It helps estimate resources, cost, effort, schedule, and risks.

Planning activities

Activity	Meaning
Define scope	Decide what is included and excluded in the project.
Check feasibility	Check whether the project is possible technically and financially.
Analyze risks	Find possible problems before development.
Define resources	Identify people, tools, hardware, and reusable software.
Estimate cost and effort	Calculate required work, time, and money.
Prepare schedule	Decide start and finish dates for tasks.
Update plan	Change the plan when requirements or conditions change.

Important: Planning is continuous and must be updated during the project.

3. Software Project Estimation

Estimation predicts the effort, time, cost, and resources required to complete a software project.

Key components

Component	Meaning
Time	Duration required for the project.
Cost	Money required for the project.
Resources	People, tools, hardware, and software.
Effort	Total human work, measured in person-months.

Estimation methods

Method	Meaning
Delay estimation	Wait until more project details are available.
Similar-project estimation	Use data from previous similar projects.
Decomposition	Break the project into small modules and estimate each.
Mathematical models	Use formulas such as COCOMO to estimate effort and cost.

4. Software Sizing and LOC

Software sizing measures the amount of software to be developed. LOC measures the number of source-code lines.

LOC formulas

Effort = LOC / Productivity
Cost = Effort x Cost per person-month
Schedule = Effort / Number of developers

Example

Given	Calculation	Result
25,000 LOC, 600 LOC/person-month	25,000 / 600	42 person-months
5 developers	42 / 5	8.4 months
$6,000 per person-month	42 x 6,000	$252,000

LOC advantage: Simple and useful when design/code is known. **LOC limitation:** Not suitable in early stages or for new technology.

5. Function Point Metrics

A Function Point measures the functionality delivered to the user. It does not depend directly on programming language or code lines.

Five FP components

Component	Meaning	Example
EI	External Input: data enters the system.	Login form, payment details.
EO	External Output: system gives information to user.	Reports, receipts, error messages.
EQ	External Inquiry: input gives immediate response.	Product search result.
ILF	Internal Logical File: data stored inside application.	User or order database.
EIF	External Interface File: data used from outside application.	Bank data used by payment app.

FP formula

FP = Count Total x [0.65 + 0.01 x Sum(Fi)]
Example: FP = 50 x [0.65 + 0.01 x 17] = 41 Function Points.

Exam point: LOC measures code written; FP measures functionality delivered.

6. COCOMO Model

COCOMO means Constructive Cost Estimation Model. It was proposed by Barry Boehm and estimates effort, development time, cost, and team size using KLOC.

Formulas

Effort = a1 x (KLOC)^a2
Development Time = b1 x (Effort)^b2

Project types and constants

Project type	Meaning	a1	a2	b1	b2
Organic	Small, simple, experienced team.	2.4	1.05	2.5	0.38
Semidetached	Medium size and medium complexity.	3.0	1.12	2.5	0.35
Embedded	Large, complex, strict constraints.	3.6	1.20	2.5	0.32

Three COCOMO models

- **Basic COCOMO:** Uses KLOC and project type.
- **Intermediate COCOMO:** Uses KLOC, project type, and cost drivers.
- **Complete COCOMO:** Considers each software life-cycle phase separately.

Important: KLOC means 1,000 lines of code. Effort is measured in person-months (PM).

7. Project Scheduling and Tracking

Scheduling distributes project effort across the planned duration and assigns tasks to team members with start and finish dates.

Scheduling principles

Principle	Meaning
Compartmentalization	Divide project into manageable tasks.
Interdependency	Identify parallel and serial tasks.
Time allocation	Give start and finish dates to tasks.
Effort validation	Ensure enough staff are available.
Responsibilities	Assign every task to a person.
Outcomes	Every task must produce a deliverable.
Milestones	Important completed achievements after quality review.

40-20-40 rule

Activity	Effort
Analysis and design	40% or more
Programming	20%
Testing	40%

8. PERT, CPM and Gantt Chart

PERT means Program Evaluation and Review Technique. CPM means Critical Path Method. Both help schedule tasks and find the project duration.

Important points

- **PERT:** Estimates task time and project duration using uncertainty.
- **CPM:** Finds the critical path and task dependencies.
- **Critical path:** Longest chain of dependent tasks; it decides project duration.
- **Gantt chart:** Shows tasks against time using horizontal bars.

Gantt chart example

Task	Month 1	Month 2	Month 3	Month 4
Requirements	Yes			
Design		Yes		
Coding		Yes	Yes	
Testing			Yes	
Release				Yes

Schedule tracking methods

- Conduct regular status meetings.
- Check completed milestones.
- Compare actual and planned dates.
- Review testing and development results.
- Use Earned Value Analysis for progress measurement.

9. Risk Analysis and Management

A risk is a possible future problem that may or may not happen. If it occurs, it may cause delay, extra cost, or loss.

Risk types

Risk type	Effect
Project risk	Affects project plan, schedule, cost, and resources.
Technical risk	Affects software quality, performance, and implementation.
Business risk	Affects product feasibility and business success.

Business risk categories

- **Market risk:** Product is good but nobody wants it.
- **Strategic risk:** Product does not match company strategy.
- **Sales risk:** Sales team cannot sell the product.
- **Management risk:** Senior management support is lost.
- **Budget risk:** Money or staff commitment is reduced.

Risk management steps

- **Risk identification:** Find possible risks.
- **Risk analysis:** Estimate probability and impact.
- **Risk ranking:** Give priority to high-probability and high-impact risks.
- **Contingency planning:** Prepare an emergency plan.

RMMM

Part	Meaning
Mitigation	Avoid or reduce the risk before it occurs.

Part	Meaning
Monitoring	Track warning signs and check risk level.
Management	Use contingency plan after risk occurs.

RIS means Risk Information Sheet. It records risk description, probability, impact, priority, mitigation, monitoring, and contingency plan.

Quick Revision

Topic	Remember
Project manager	Controls time, cost, quality, resources, and risks.
Planning	Creates project roadmap and estimates.
Estimation	Predicts effort, time, cost, and resources.
LOC	Measures lines of code.
Function Point	Measures user functionality.
COCOMO	Estimates effort and time using KLOC.
Scheduling	Assigns tasks, people, and dates.
40-20-40 rule	40% design, 20% coding, 40% testing.
PERT	Estimates project duration.
CPM	Finds critical path.
Gantt chart	Shows tasks against time.
Risk	Possible future problem.
RMMM	Mitigation, Monitoring, and Management.
RIS	Risk Information Sheet.

Most important formulas

Effort = LOC / Productivity
FP = Count Total x [0.65 + 0.01 x Sum(Fi)]
Effort = a1 x (KLOC)^a2
Development Time = b1 x (Effort)^b2